

Final Report: Minimal Conformal Non-Abelian Model for a PTA-Scale Signal

DarkAgents

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1 Summary

The completed `fopt-pta` pipeline identifies the minimal conformal non-Abelian candidate as a classically scale-invariant dark $SU(2)_D$ gauge theory with one complex scalar doublet. The local authoritative model uses a zero-portal secluded dark sector, two free parameters (g_D, v_D) , and the vector-mass convention $m_{W_D}(\chi) = g_D \chi / 2$.

The exact core model is not new: Borah, Dasgupta and Kang studied the same classically conformal dark $SU(2)_D$ doublet model as a NANOGrav 12.5-year first-order phase-transition interpretation in Ref. [1]. The present pipeline should therefore be read as a local recomputation and updated PTA-likelihood exercise, not as a model-level novelty claim.

The first-order phase-transition (FOPT) scan found many viable points. The PTA benchmark scan found a near-match to the PTA violin windows, with best score 13/14. The PTArcade campaign completed with status `ok`, $N_{\text{samples}} = 100000$, and maximum-a-posteriori (MAP) point

$$g_D = 0.863175, \quad v_D = 1.10802 \text{ GeV}.$$

This is a conditional result. The dominant limitations are the DBF gravitational-wave template validity at extremely large α , the absence of an independent wall-velocity or efficiency calculation, and missing thermal-history, BBN/CMB, ΔN_{eff} , relic-density, and self-interaction observables. The constraints stage found no direct exclusion, but mostly because the necessary observables are absent, not because full cosmological viability has been proven.

2 Prompt

what is the minimal conformal non-abelian model that can explain the NANOGrav signal?

3 Model Description

The critic-authoritative model is `minimal_conformal_su2_doublet`. It contains a dark non-Abelian gauge group $SU(2)_D$ with coupling g_D and one complex scalar doublet Φ , while all new fields are Standard-Model singlets. Classical scale invariance is imposed at tree level, forbidding explicit scalar mass terms. In unitary gauge,

$$\Phi = (0) \chi / \sqrt{2}, \quad \langle \chi \rangle = v_D.$$

The tree-level scalar potential is

$$V_0(\chi) = \frac{\lambda_\phi}{4} \chi^4,$$

where λ_ϕ is Coleman-Weinberg-fixed rather than an independent scan parameter. The symmetry-breaking pattern is

$$SU(2)_D \rightarrow \emptyset,$$

so the three dark vectors become massive, with

$$m_{W_D}^2(\chi) = \frac{g_D^2 \chi^2}{4}.$$

The free parameters used downstream are $g_D > 0$ and $v_D > 0$. The local handoff flags a convention tension with Ref. [1]: that paper states a physical vector mass convention consistent with $M_{Z_D} = g_D M/2$, while also displaying a field-dependent mass normalization that is not directly identical to the local convention. All local computations use the critic-authoritative convention above.

4 Pipeline Status

Stage	Status	Interpretation
Model / critique	warning	Usable model; no red flags, but same-model precedent and mass-convention caveat.
Preliminary literature	warning	Exact core model already studied in Ref. [1]; benchmark conventions require care.
FOPT scan	ok	934 points scanned, 662 viable, 268 failed, 4 excluded.
PTA analysis	warning	Benchmark and PTArcade completed, but no 14/14 benchmark and template/prior caveats remain.
Constraints	warning	No exclusion claimed; most relevant observables require new backend calculations.
Prior audit	warning	High-severity limitations dominate the final interpretation.

5 Preliminary Literature Search

The same-model precedent is Ref. [1], which studied a classically conformal dark $SU(2)_D$ model with one complex scalar doublet and a negligible Higgs portal for a NANOGrav 12.5-year FOPT interpretation. The directly quoted benchmark from that work is $g_D = 1.37$, $M_{Z_D} = 8.23$ MeV, $T_\star = 1.8$ MeV, $\alpha_\star = 0.45$, and $\phi_c/T_c = 3.56$. This benchmark should not be imported into the local parameter basis without resolving the vector-mass convention.

Several structurally related works provide context rather than direct benchmarks. Classically conformal Abelian hidden-sector models were studied for nano-Hertz gravitational waves in Refs. [2, 4]. Nearly conformal or phenomenological dark-sector phase-transition explanations of NANOGrav were considered in Refs. [3, 5]. These papers support the plausibility of MeV-to-sub-GeV dark-sector FOPTs as PTA-scale sources, but they are not the exact non-Abelian scalar-doublet model analyzed here.

6 FOPT Results

The FOPT agent used a semianalytic conformal high-temperature/polynomial backend with percolation and perturbativity checks. The scan ranges were

$$0.25 \leq g_D \leq 3.2, \quad 10^{-5} \text{ GeV} \leq v_D \leq 0.8 \text{ GeV}.$$

Out of 934 scanned points, 662 were viable, 268 failed, and 4 were excluded by validation criteria. Representative high-ranking steering points are shown in Table 1.

Table 1: Representative FOPT steering points. Frequencies are order-of-magnitude steering estimates, not template-derived spectral peaks.

Point	g_D	v_D [GeV]	α	β/H	T_n [GeV]	f_{est} [Hz]
point_0594	0.900	4.20×10^{-2}	2.10×10^{10}	21.21	1.56×10^{-5}	1.00×10^{-8}
point_0769	1.400	4.11×10^{-3}	1.04×10^1	148.63	3.30×10^{-4}	9.92×10^{-9}
point_0681	1.150	1.20×10^{-2}	9.30×10^2	62.18	2.86×10^{-4}	9.90×10^{-9}
point_0441	1.823	1.04×10^{-3}	8.65×10^{-1}	381.21	2.00×10^{-4}	1.00×10^{-8}
point_0715	1.250	7.03×10^{-3}	7.25×10^1	97.70	3.21×10^{-4}	9.81×10^{-9}

The FOPT result is encouraging as a PTA-band steering calculation, but it is not a full finite-temperature effective-potential and bounce validation. The inherited caveats are the high-temperature approximation, Gaussian percolation approximation, lack of quantified gauge/daisy/RG uncertainty, and no independent wall-dynamics calculation.

7 PTA Results

The PTA stage used a benchmark scan plus a PTArcade campaign. For the benchmark comparison, the selected implemented spectral template was DBF. The scan considered 2611 points, of which 2546 were viable, and the best direct benchmark scored 13/14 against the PTA violin windows. The best benchmark point was

$$g_D = 0.95, \quad v_D = 0.360438 \text{ GeV}, \quad \alpha = 3.73 \times 10^8, \quad \beta/H = 23.69, \quad T_\star = 0.03237 \text{ GeV}.$$

Its peak grid point was near 10^{-8} Hz with $h^2\Omega_{\text{GW}} \simeq 2.83 \times 10^{-9}$.

The PTArcade run status was `ok`. The campaign used the NANOGrav 15-year `ceffyl` mode with priors

$$g_D \sim U(0.45, 1.55), \quad \log_{10}(v_D/\text{GeV}) \sim U(-3.2, 0.7),$$

and fixed DBF template. The accepted chain contains $N_{\text{samples}} = 100000$. The posterior mean values are

$$g_D = 0.8836 \pm 0.0353, \quad \log_{10}(v_D/\text{GeV}) = -0.0539 \pm 0.1716,$$

with 68% credible intervals $g_D \in [0.8528, 0.9159]$ and $\log_{10}(v_D/\text{GeV}) \in [-0.2217, 0.1005]$. The MAP point is

$$g_D = 0.863175, \quad \log_{10}(v_D/\text{GeV}) = 0.044548, \quad v_D = 1.10802 \text{ GeV},$$

$$T_{\text{reh}} = 0.09782 \text{ GeV}, \quad \alpha = 2.14 \times 10^{14}, \quad \beta/H = 13.98, \quad f_{\text{peak,est}} = 1.75 \times 10^{-7} \text{ Hz}.$$

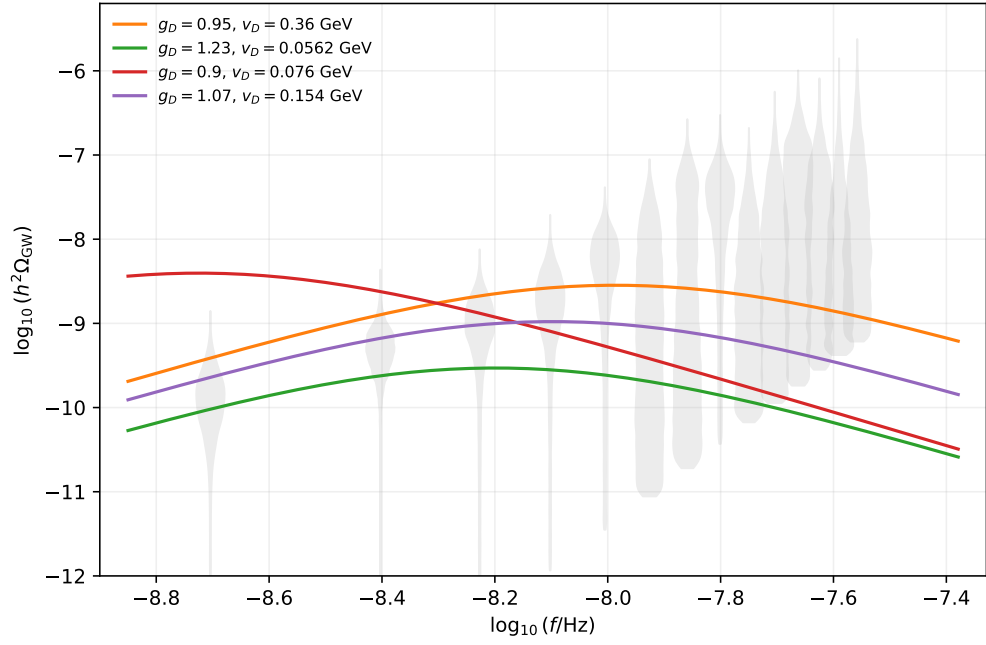


Figure 1: Benchmark DBF spectrum compared with PTA violin windows. The best benchmark score is 13/14.

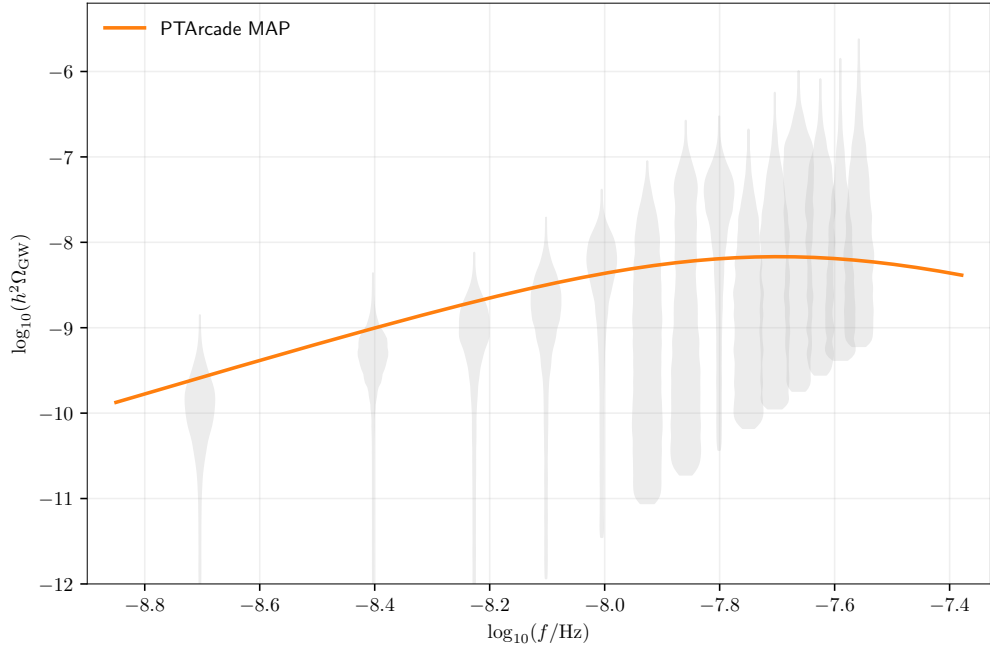


Figure 2: PTArcade MAP DBF spectrum for the accepted widened-prior run.

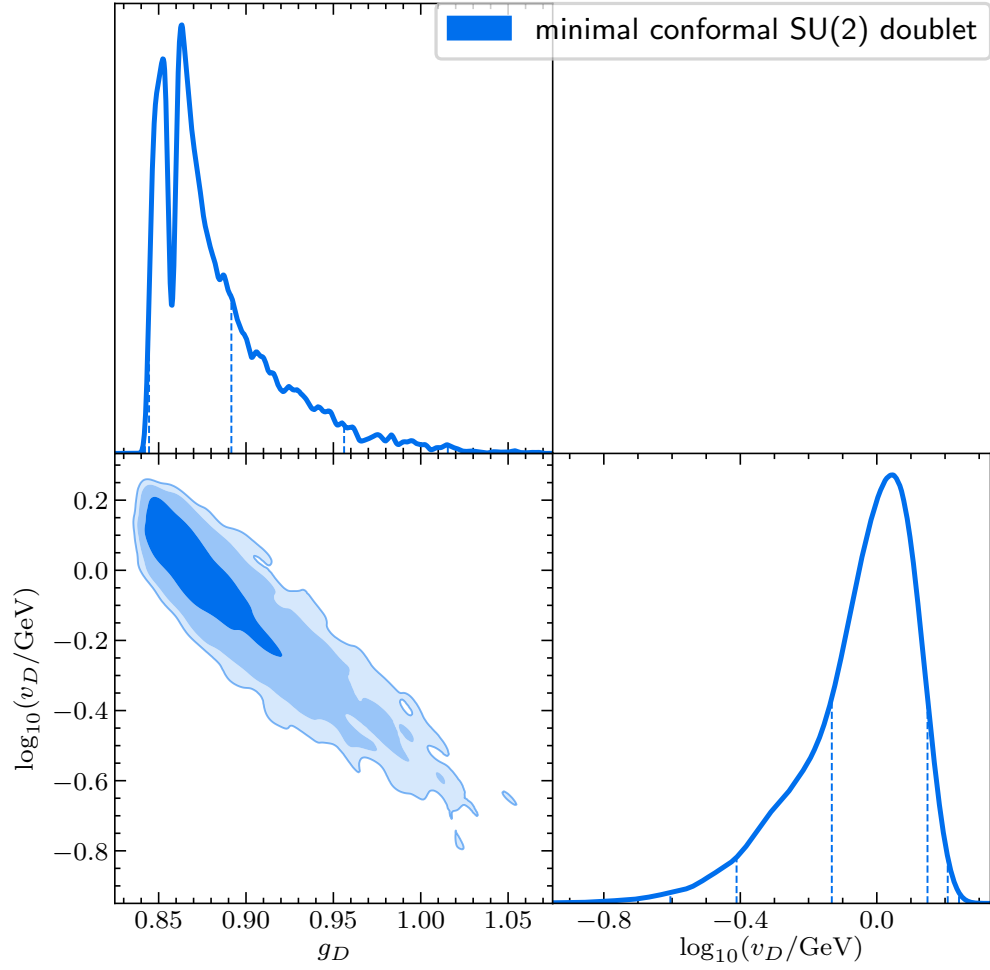


Figure 3: PTArcade posterior distributions for the fitted parameter basis.

The MAP $f_{\text{peak,est}}$ is an approximate steering diagnostic and lies above the nominal PTA steering band; the fitted object is the DBF spectrum, not this scalar diagnostic alone.

The main PTA caveat is template validity. The DBF template was selected as the closest implemented template for very large α , while the Higgsless template was not selected because many local points exceed its stated $\alpha \lesssim 0.5$ validation hint. DBF amplitudes use backend defaults, including efficiency assumptions, while wall velocity and microscopic efficiency are not computed independently.

8 Constraints

The constraints agent considered 12 constraint classes: 1 directly testable theory check, 7 requiring new backend calculations, and 4 not applicable to the exact zero-portal model. The directly testable perturbativity check passed for the posterior region, with g_D well below $\sqrt{4\pi}$. This is a basic consistency check, not a complete theory-validity proof.

Table 2: Constraint classification. “No exclusion” means no exclusion with the observables currently available.

Class	Examples	Applicability	Missing ingredient
BBN/CMB thermal dark sectors	Refs. [6, 7]	Requires new back- end	T_D/T_{SM} , abundance, en- tropy transfer, likelihood map.
ΔN_{eff} and dark radiation	Planck [8]	Requires new back- end	Decoupling history and ra- diation budget.
PTA-scale en- ergy release	Ref. [5]	Requires new back- end	Visible/dark reheating par- tition and late energy injec- tion.
Self-interacting DM	Ref. [9]; Bullet Clus- ter [10]	Requires new back- end	$\sigma_T/m(v)$ and dark-vector DM fraction.
Portal labora- tory searches	Refs. [11, 12, 13]	Not applicable	Zero portal in the minimal model; relevant again for portal extensions.
Perturbativity	Upstream handoffs	Direct	Passed basic $g_D < \sqrt{4\pi}$ check; higher-loop validity not tested.

The most important missing backend calculations are a thermal-history module, BBN/CMB observable mapping, vector relic-density calculation, non-Abelian vector self-interaction cross sections, portal widths for non-minimal variants, and higher-order/RG validity diagnostics.

9 Assumptions and Limitations

The prior audit identified 18 assumptions, limitations, priors, and caveats. The highest-severity items are:

- No thermal history, dark-sector temperature ratio, decoupling history, reheating partition, relic abundance, or entropy-transfer observable was computed.

- Sub-GeV strongly supercooled FOPTs can affect BBN/CMB through energy release or dark radiation, but ΔN_{eff} and visible/dark reheating were not mapped.
- The minimal model fixes the Higgs portal to zero, which removes direct portal searches but leaves the origin and thermal population of the dark sector unspecified.
- No dark-vector relic abundance or fractional dark-matter abundance was computed, so dark-matter viability and self-interaction constraints cannot be applied.
- The PTA analysis uses DBF spectra at extremely large α without an independent calculation of wall velocity or microscopic efficiency.
- The FOPT calculation is semianalytic and does not quantify gauge, daisy, RG-scale, or full-bounce uncertainties.

Consequently, the PTA fit should be interpreted as a conditional spectrum-level match within the implemented backend, not as a complete cosmological viability proof.

10 Conclusions

Robust claims are limited to the structure of the local candidate and the executed pipeline outputs. The minimal conformal non-Abelian candidate is a dark $SU(2)_D$ gauge theory with one complex scalar doublet, classically scale-invariant tree potential, and two local free parameters (g_D, v_D) . The exact core model has same-model precedent in Ref. [1]. The local FOPT scan found many viable points, and the PTA workflow found a strong benchmark near-match and an `ok` PTArcade posterior run.

Conditional claims are the PTA interpretation and parameter preference. They depend on the DBF template, backend approximations, chosen priors, and missing wall-dynamics inputs. The reported MAP point is a valid output of the local PTArcade campaign, but its extremely large α and approximate peak diagnostic require caution.

Blocked claims are cosmological viability, dark-matter viability, and self-interaction safety. These require new thermal-history, BBN/CMB, ΔN_{eff} , relic-density, and self-interaction backends.

Uncontrolled claims would include saying that the model is novel, proven viable, or excluded-free in a full cosmological sense. The current constraints result is better phrased as: no exclusion was derived from the available observables, and most relevant observables remain missing.

References

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